

# Orthonormal Bases

Def<sup>n</sup> (Orthonormal)

A set of vectors is called orthonormal if each vector in the set has norm 1 and is orthogonal to all other vectors in the set.

$S = \{\bar{v}_1, \bar{v}_2, \dots, \bar{v}_n\} \subset V$ , then  $S$  is called set of ~~orthogonal~~ orthonormal vectors if

$$\langle \bar{v}_i, \bar{v}_j \rangle = \delta_{ij} \quad \forall i, j \in \{1, \dots, n\}.$$

Example: standard basis in  $F^n$  is an orthonormal set.

lemma. If  $\{\bar{e}_1, \bar{e}_2, \dots, \bar{e}_n\}_n$  is an orthonormal set in  $V$ , then

$$\left\| \sum_{i=1}^n a_i \bar{e}_i \right\|^2 = \sum_{i=1}^n |a_i|^2 \quad \forall a_i \in F, i \in \{1, \dots, n\}.$$

lemma. An orthonormal set of vectors is linearly indep.

Proof:  $\{\bar{e}_1, \dots, \bar{e}_n\}$  be orthonormal set in  $V$ .

$$\sum_{i=1}^n a_i \bar{e}_i = 0 \Rightarrow \left\| \sum_{i=1}^n a_i \bar{e}_i \right\| = 0 \Rightarrow \sum_{i=1}^n |a_i|^2 = 0$$

$\forall i \in \{1, \dots, n\}, a_i = 0.$

Def<sup>n</sup> (Orthonormal basis)

An orthonormal set of vectors that spans the vector space is called an orthonormal basis.

Corollary: Any orthonormal set of vectors w/  $\dim V$  elements forms an orthonormal basis of  $V$ .

Thm. Let  $\{\bar{e}_1, \dots, \bar{e}_n\}$  be an orthonormal basis of  $V$  and  $\bar{v} \in V$ . Then

$$\bar{v} = \langle \bar{v}, \bar{e}_1 \rangle \bar{e}_1 + \dots + \langle \bar{v}, \bar{e}_n \rangle \bar{e}_n = \sum_{i=1}^n \langle \bar{v}, \bar{e}_i \rangle \bar{e}_i$$

and  $\|\bar{v}\|^2 = \sum_{i=1}^n |\langle \bar{v}, \bar{e}_i \rangle|^2$ .

Proof: Say  $\bar{v} = \sum_{i=1}^n a_i \bar{e}_i$ , then  $a_i = \langle \bar{v}, \bar{e}_i \rangle$   $\forall i \in \{1, \dots, n\}$ .

$$\|\bar{v}\|^2 = \sum_{i=1}^n |a_i|^2 = \sum_{i=1}^n |\langle \bar{v}, \bar{e}_i \rangle|^2$$

✓ Gram-Schmidt procedure gives a method for turning a linearly independent set into an orthonormal set with same span as the original set.

Thm (Gram-Schmidt Procedure) Let  $\{v_1, \dots, v_n\}$  be linearly independent <sup>set of</sup> vectors in  $V$ . Let  $\bar{e}_1 = \frac{\bar{v}_1}{\|\bar{v}_1\|}$ .

For  $j \in \{2, \dots, n\}$ , define  $\bar{e}_j$  inductively by

$$\bar{e}_j = \frac{\bar{v}_j - \sum_{i=2}^j \langle \bar{v}_j, \bar{e}_{i-1} \rangle \bar{e}_{i-1}}{\|\bar{v}_j - \sum_{i=2}^j \langle \bar{v}_j, \bar{e}_{i-1} \rangle \bar{e}_{i-1}\|}$$

Then  $\{\bar{e}_1, \dots, \bar{e}_n\}$  is an orthonormal set s.t.

$$\text{span}\{\bar{v}_1, \dots, \bar{v}_j\} = \text{span}\{\bar{e}_1, \dots, \bar{e}_j\}$$

$\forall j \in \{1, \dots, n\}$ .



Proof. By induction on  $j$ .

For  $j=1$ ,  $\text{span}(\bar{v}_1) = \text{span}(\bar{e}_1)$ ,  $\bar{e}_1 = \frac{1}{\|\bar{v}_1\|} \bar{v}_1$ .

Let  $1 < j < \infty$  and  $\text{span}(\bar{v}_1, \dots, \bar{v}_{j-1}) = \text{span}(\bar{e}_1, \dots, \bar{e}_{j-1})$ .

$\bar{v}_j \notin \text{span}(\bar{v}_1, \dots, \bar{v}_{j-1}) \Rightarrow \bar{v}_j \notin \text{span}(\bar{e}_1, \dots, \bar{e}_{j-1})$ .  
 $\therefore \bar{e}_j$  is well-defined and  $\|\bar{e}_j\| = 1$ .

Let  $1 \leq k \leq j$ , then

$$\langle \bar{e}_j, \bar{e}_k \rangle = \left\langle \frac{\bar{v}_j - \sum_{i=2}^j \langle \bar{v}_j, \bar{e}_{i-1} \rangle \bar{e}_{i-1}}{\|\bar{v}_j - \sum_{i=2}^j \langle \bar{v}_j, \bar{e}_{i-1} \rangle \bar{e}_{i-1}\|}, \bar{e}_k \right\rangle$$

$$= \frac{\langle \bar{v}_j, \bar{e}_k \rangle - \sum_{i=2}^j \langle \bar{v}_j, \bar{e}_{i-1} \rangle \langle \bar{e}_{i-1}, \bar{e}_k \rangle}{\|\bar{v}_j - \sum_{i=2}^j \langle \bar{v}_j, \bar{e}_{i-1} \rangle \bar{e}_{i-1}\|} = 0$$

$\Rightarrow \{\bar{e}_1, \dots, \bar{e}_j\}$  is an orthonormal set.

From def<sup>n</sup> of  $\bar{e}_j$ ,  $\bar{v}_j \in \text{span}(\bar{e}_1, \dots, \bar{e}_j)$ .

We have,  $\text{span}(\bar{v}_1, \dots, \bar{v}_j) \subset \text{span}(\bar{e}_1, \dots, \bar{e}_j)$ .

Both sets are linearly independent & spans vectors in both.  $\therefore$  Both sets are equal.

Thm. Every finite-dim vector space w/ an inner product ~~space~~ has an orthonormal basis.

That is every finite-dim inner product space has an orthonormal basis.

Proof: Consider a basis & construct an orthonormal basis w/ Gram Schmidt procedure.

Thm. Every orthonormal set in a finite-dim vector space can be extended to an orthonormal basis. (Note, vector space is inner product space).

Example: Consider  $\mathcal{P}_2(\mathbb{R})$  w/  $\langle p, q \rangle = \int_{-1}^1 p(x) q(x) dx$ .  
Consider a basis  $\{1, x, x^2\}$ .

$$\|1\|^2 = \int_{-1}^1 1^2 dx = 2. \text{ Then, } \bar{e}_1 = \sqrt{\frac{1}{2}} x^0.$$

$$\text{To find } \bar{e}_2, \quad x - \langle x, \bar{e}_1 \rangle \bar{e}_1 = x - \left( \int_{-1}^1 x \sqrt{\frac{1}{2}} dx \right) \sqrt{\frac{1}{2}} = x.$$

$$\|x\|^2 = \frac{2}{3}. \text{ Then, } \bar{e}_2 = \sqrt{\frac{3}{2}} x.$$

To find  $\bar{e}_3$ ,

$$\begin{aligned} & x^2 - \langle x^2, \bar{e}_1 \rangle \bar{e}_1 - \langle x^2, \bar{e}_2 \rangle \bar{e}_2 \\ &= x^2 - \left( \int_{-1}^1 x^2 \sqrt{\frac{1}{2}} dx \right) \sqrt{\frac{1}{2}} - \left( \int_{-1}^1 x^2 \sqrt{\frac{3}{2}} x dx \right) \sqrt{\frac{3}{2}} x \\ &= x^2 - \frac{1}{3}. \quad \therefore \|x^2 - \frac{1}{3}\|^2 = \frac{8}{45}. \end{aligned}$$

$$\text{Then, } \bar{e}_3 = \sqrt{\frac{45}{8}} \left( x^2 - \frac{1}{3} \right).$$

$\{\bar{e}_1, \bar{e}_2, \bar{e}_3\}$  is an orthonormal basis of  $\mathcal{P}_2(\mathbb{R})$ .

Thm. Suppose  $T \in L(V)$ . If  $T$  has an upper triangular matrix w/ some basis of  $V$ , then  $T$  has an upper triangular w/ some orthonormal basis of  $V$ .

Proof.



## Thm. (Schur's theorem)

Let  $V$  be a finite-dim. complex vector space &  $T \in L(V)$ . Then  $T$  has an upper triangular matrix w/ some orthonormal basis of  $V$ .

Linear functionals in inner product spaces.

Defn A linear functional on  $V$  is a linear map from  $V$  to  $F$ .

That is, a linear functional is an element of  $L(V, F)$ .

Example: Inner product  $\langle \cdot, \bar{u} \rangle$  w/ respect to a fixed  $\bar{u} \in V$  is a linear functional.

## Thm. (Riesz representation theorem)

Let  $V$  be a finite-dim. &  $\varphi: V \rightarrow F$  be a linear functional. Then  $\exists$  a unique  $\bar{u} \in V$  s.t.  
 $\varphi(\bar{v}) = \langle \bar{v}, \bar{u} \rangle \quad \forall \bar{v} \in V$ .

Proof! 1<sup>st</sup> we show  $\exists \bar{u} \in V$  s.t.  $\varphi(\bar{v}) = \langle \bar{v}, \bar{u} \rangle \quad \forall \bar{v} \in V$ .  
Let  $\{\bar{e}_1, \dots, \bar{e}_n\}$  be basis of  $V$  that's orthonormal. Then,

$$\begin{aligned}\varphi(\bar{v}) &= \varphi(\langle \bar{v}, \bar{e}_1 \rangle \bar{e}_1 + \dots + \langle \bar{v}, \bar{e}_n \rangle \bar{e}_n) \\ &= \langle \bar{v}, \overline{\varphi(\bar{e}_1)} \bar{e}_1 + \dots + \overline{\varphi(\bar{e}_n)} \bar{e}_n \rangle \quad \forall \bar{v} \in V.\end{aligned}$$

$$\text{That is, } \bar{u} = \sum \overline{\varphi(\bar{e}_i)} \bar{e}_i, \quad \varphi(\bar{v}) = \langle \bar{v}, \bar{u} \rangle \quad \forall \bar{v} \in V.$$

Suppose,  $\varphi(\bar{v}) = \langle \bar{v}, \bar{u}_1 \rangle = \langle \bar{v}, \bar{u}_2 \rangle \quad \forall \bar{v} \in V$ . Then  
 $0 = \langle \bar{v}, \bar{u}_1 \rangle - \langle \bar{v}, \bar{u}_2 \rangle = \langle \bar{v}, \bar{u}_1 - \bar{u}_2 \rangle \quad \forall \bar{v} \in V$ , ~~then~~ &  
for  $\bar{v} = \bar{u}_1 - \bar{u}_2 \Rightarrow \|\bar{u}_1 - \bar{u}_2\|^2 = 0 \Rightarrow \bar{u}_1 = \bar{u}_2$ .